OPTICS · SEMESTER 5 · MODULE 3

Answer key: Diffraction

Formal solutions to all 42 questions of the model paper.

BEFORE YOU READ AN ANSWER

Each answer starts with **Know first**: topic codes from File 1. If one feels new, read it first, try the question yourself, then read the solution.

Grey mark notes show a typical split of marks (a guide, not an official scheme).

Section A · Short answers

2 marks each.

1 PYQ N25 Q5

Fresnel vs Fraunhofer diffraction.

KNOW FIRST

T1 Diffraction; Fresnel and Fraunhofer P2 Huygens' principle and wavefronts

Fresnel	Source and/or screen at finite distances from the aperture; incident wavefronts are spherical or cylindrical; no lenses needed; the pattern depends on the distances.
Fraunhofer	Source and screen effectively at infinity; plane wavefronts fall on the aperture (realised with two convex lenses); the pattern is in the focal plane of the second lens.

N20 Q4 (fill in the blank: "source or screen or both at finite distances"): **Fresnel**.

2

Huygens–Fresnel principle.

KNOW FIRST

T1 Diffraction; Fresnel and Fraunhofer P2 Huygens' principle and wavefronts

Every point on a wavefront acts as a source of secondary spherical wavelets, and the wavelets from different points mutually interfere. The disturbance at any point beyond is obtained by superposing (with their phases) all the secondary wavelets reaching that point.

3

Interference vs diffraction.

KNOW FIRST

T1 Diffraction; Fresnel and Fraunhofer T5 Two slits: fringes inside an envelope

Both are superposition of waves; there is no fundamental difference. Interference refers to superposition of waves from a few separate coherent sources (e.g. two slits), giving fringes of equal width and nearly equal intensity. Diffraction refers to superposition of wavelets from a continuous set of points of the same wavefront (one aperture), giving a pattern with a bright central maximum and rapidly weakening, unequal side bands.

4

Effect of slit width and wavelength.

KNOW FIRST

T3 Single slit: dark bands, weak bright bands, width

The minima are at $b\sin\theta = m\lambda$, so the central maximum has angular half-width λ/b. (a) A narrower slit (smaller b) gives a **wider** pattern, with the central maximum spreading out (and getting fainter). (b) A longer wavelength also widens the pattern: red light gives a wider pattern than blue.

5

Positions of secondary maxima.

KNOW FIRST

T3 Single slit: dark bands, weak bright bands, width

M4 Solving tan β = β with a graph

Setting $dI/d\beta = 0$ for $I = I_0 \sin^2\beta/\beta^2$ gives $\sin\beta(\beta - \tan\beta) = 0$. The maxima satisfy $\tan\beta = \beta$, whose roots are $\beta = 1.43\pi, 2.46\pi, 3.47\pi, \ldots$, slightly less than $1.5\pi, 2.5\pi, \ldots$. They are not midway because the factor $1/\beta^2$ falls off with β, pulling each maximum towards the centre. Intensities are about 4.7%, 1.6%, 0.8% of I_0.

6

Sound bends round buildings, light does not.

KNOW FIRST

T1 Diffraction; Fresnel and Fraunhofer

The spreading by diffraction is of angular size about λ/b, where b is the size of the opening or obstacle. Sound wavelengths (about 1 m) are comparable with doors and buildings, so diffraction is large. Light wavelengths (about 5×10^{-7} m) are millions of times smaller than everyday objects, so the bending is negligible and light appears to travel in straight lines.

7 PYQ N19 Q7 · adapted

Circular aperture gives concentric rings?

KNOW FIRST

T4 A round hole, and the limit of resolution

True. Fraunhofer diffraction by a circular aperture gives a bright central disc (Airy disc) surrounded by alternate dark and bright concentric rings of rapidly decreasing intensity. The first dark ring is at $\sin\theta = 1.22\lambda/D$.

8 PYQ N19 Q12

Loudspeaker diameter for directionality.

KNOW FIRST

T4 A round hole, and the limit of resolution T1 Diffraction; Fresnel and Fraunhofer

Sound from the speaker spreads by diffraction over an angle $\theta \approx 1.22\lambda/D$ (D = diameter). To get a narrow, directional beam we need small θ, so the diameter should be **large** compared with the wavelength of the sound. (That is why high-frequency tweeters, with short λ, can be small, while directional speakers for low frequencies are large.)

9

Missing orders.

KNOW FIRST

T5 Two slits: fringes inside an envelope

In the double-slit pattern, interference maxima occur at $d\sin\theta = m\lambda$ and diffraction minima at $b\sin\theta = p\lambda$. If both hold for the same θ, the interference maximum falls on a diffraction zero and is absent: a missing order. Condition: $\frac{d}{b} = \frac{m}{p}$. E.g. $d = 2b$: orders 2, 4, 6, ... are missing.

10 PYQ N22 Q10 · adapted

Principal maximum sharper as N increases?

KNOW FIRST

T6 N slits: the intensity distribution T7 Width of a principal maximum

True. The angular half-width of a principal maximum is $\Delta\theta = \dfrac{\lambda}{Nd\cos\theta}$, which decreases as N increases, while the peak intensity grows as N^2. (So the N19 Q6 statement "sharper as N decreases" is false.)

11

Minima and secondary maxima between principal maxima; N² intensity.

KNOW FIRST

T6 N slits: the intensity distribution

Minima occur at $N\gamma = p\pi$ with $p \neq 0, N, 2N, \ldots$, so between two adjacent principal maxima there are $N-1$ minima and hence $N-2$ secondary maxima. At a principal maximum ($d\sin\theta = m\lambda$) the waves from all N slits arrive in phase, so the amplitudes add to N times one slit's; intensity ∝ amplitude² ∝ N^2.

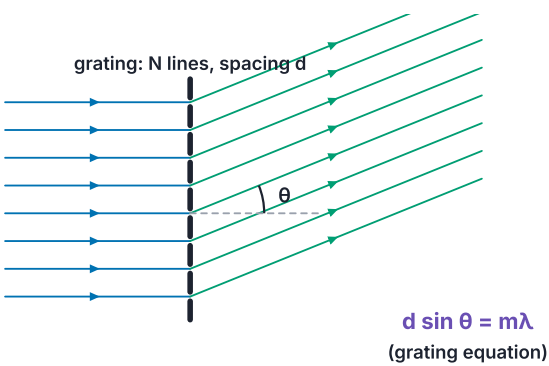
12 PYQ N20 Q15

Explain a diffraction grating.

KNOW FIRST

T8 The diffraction grating

A diffraction grating is a large number of equidistant, parallel, narrow slits of equal width, usually made by ruling fine lines (several thousand per cm) on glass. The distance d between corresponding points of neighbouring slits is the grating element (= 1 ÷ number of lines per unit length). Light of wavelength λ gives sharp principal maxima in directions $d \sin\theta = m\lambda$, so a grating disperses light into spectra of different orders and is used to measure wavelengths.



A diffraction grating: many equally spaced slits. Bright lines appear only in directions where light from neighbouring slits differs in path by a whole number of wavelengths.

13 PYQ N19 Q9 · adapted

Larger grating element, larger dispersion?

KNOW FIRST

T8 The diffraction grating

False. Angular dispersion $\frac{d\theta}{d\lambda} = \frac{m}{d\cos\theta}$ is inversely proportional to the grating element d. A larger d (fewer lines per cm) gives smaller dispersion.

14 PYQ N22 Q15

Overlapping spectra.

KNOW FIRST

T8 The diffraction grating

Since $d\sin\theta = m\lambda$, the m-th order of a wavelength λ_2 falls at the same angle as the (m+1)-th order of λ_1 if $m\lambda_2 = (m+1)\lambda_1$. Then spectra of adjacent orders overlap. For visible light (400–700 nm) the first and second orders do not overlap (2 × 400 = 800 nm > 700 nm), but the second and third do (3 × 400 = 1200 < 2 × 700 = 1400 nm). Filters are used to remove the unwanted order.

15 PYQ N20 Q7 · adapted

Grating spectra purer than prism spectra?

KNOW FIRST

T8 The diffraction grating T9 Resolving power of a grating

True. A grating with many lines gives very sharp principal maxima (half-width $\lambda/Nd\cos\theta$) and high resolving power (mN), so the spectral lines are narrow and well separated; the dispersion is also nearly uniform (normal spectrum). A prism's spectrum depends on how n varies with λ, is non-linear, and has much lower resolving power.

16 PYQ N24 Q8

Equation for resolving power of a grating.

KNOW FIRST

T9 Resolving power of a grating

$$R = \frac{\lambda}{d\lambda} = mN$$

where m is the order of the spectrum and N the total number of lines illuminated. (N22 Q5 fill-in: the same equation.)

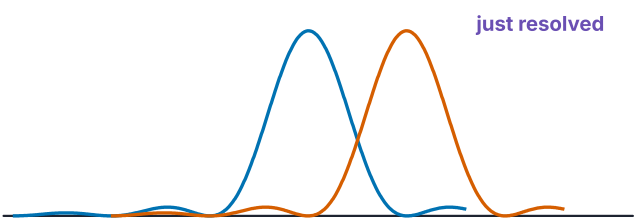
17

Rayleigh's criterion.

KNOW FIRST

T9 Resolving power of a grating T4 A round hole, and the limit of resolution

Two spectral lines (or two point objects) of equal intensity are just resolved when the principal maximum of one falls exactly on the first minimum of the other. Then the combined intensity curve shows a dip (to about 81% for slit patterns) between the two peaks, and the eye sees two lines.



Rayleigh criterion: two wavelengths are "just resolved" when the principal maximum of one falls on the first minimum of the other. For a grating this gives the resolving power $\lambda/\Delta\lambda = mN$.

18

Highest order, 5000 lines/cm, 5000 Å.

KNOW FIRST

T8 The diffraction grating M2 Small angles, limits and units

1. $d = \dfrac{1}{5000}$ cm $= 2 \times 10^{-4}$ cm; $\lambda = 5 \times 10^{-5}$ cm
2. $m_{max} = \dfrac{d}{\lambda} \sin 90^\circ = 4$

   m = 4 would need θ = 90°, i.e. light grazing along the grating: not observable.

ANSWER Highest observable order: 3

19 PYQ N25 Q6

Difference in areas of nth and (n−1)th zones.

KNOW FIRST

T10 Fresnel's half-period zones

1. $r_n^2 = \left(d + \dfrac{n\lambda}{2}\right)^2 - d^2 = n\lambda d + \dfrac{n^2\lambda^2}{4}$
2. Area of zone n: $A_n = \pi(r_n^2 - r_{n-1}^2) = \pi\lambda d + \dfrac{\pi\lambda^2}{4}(2n-1)$
3. $A_n - A_{n-1} = \dfrac{\pi\lambda^2}{4} \times 2 = \dfrac{\pi\lambda^2}{2}$

ANSWER $\pi\lambda^2/2$, negligible: to a very good approximation all zones have equal area $\pi\lambda d$

20

Whole wavefront = half the first zone.

KNOW FIRST

T10 Fresnel's half-period zones

The resultant amplitude is $u = u_1 - u_2 + u_3 - \cdots$, the u_n decreasing slowly. Write $u = \frac{u_1}{2} + \left(\frac{u_1}{2} - u_2 + \frac{u_3}{2}\right) + \left(\frac{u_3}{2} - u_4 + \frac{u_5}{2}\right) + \cdots$. Each u_n is very nearly the average of its neighbours, so each bracket is nearly zero, and the last term is negligible. Hence $u \approx u_1/2$: the whole wavefront produces half the amplitude (a quarter of the intensity) of the first zone alone.

21

Bright centre of a disc's shadow.

KNOW FIRST

T11 Straight-line travel, holes and discs T10 Fresnel's half-period zones

A small opaque disc covering the first p half-period zones (as seen from a point P on the axis) leaves the rest uncovered. Their resultant is $u_{p+1} - u_{p+2} + \cdots \approx \frac{u_{p+1}}{2}$, which is not zero; for a small disc u_{p+1} is almost as large as u_1. So the centre of the shadow is bright (Poisson's spot), as Arago observed.

22

Zone plate vs convex lens.

KNOW FIRST

T12 The zone plate

Similarity	Both focus light and obey $\frac{1}{a}+\frac{1}{b}=\frac{1}{f}$.
Principle	Zone plate: diffraction. Lens: refraction.
Foci	Zone plate: many (f, $f/3$, $f/5$, ...). Lens: one.
Colour	Zone plate $f = r_1^2/\lambda$: shorter for red. Lens: shorter for violet.
Brightness	Zone plate wastes half the light (blocked zones).

23

Intensity at the edge of the geometrical shadow.

KNOW FIRST

T13 The straight edge and the Cornu spiral

It is one quarter of the unobstructed intensity, $I_0/4$. At the edge ($v_0 = 0$) exactly half of the wavefront (one half of the Cornu spiral, from the centre O to Z^+) is effective, so the amplitude is half the unobstructed amplitude (chord OZ^+ = ½ chord Z^-Z^+), and intensity ∝ amplitude² gives $I_0/4$.

24

Cornu spiral and its end points.

KNOW FIRST

T13 The straight edge and the Cornu spiral M3 Complex numbers as arrows

The Cornu spiral is the curve traced by the point $(C(v), S(v))$, where $C(v) = \int_0^v \cos \frac{\pi u^2}{2} du$ and $S(v) = \int_0^v \sin \frac{\pi u^2}{2} du$ are Fresnel's integrals. It is a double spiral through the origin. Its end points $Z^+(\frac{1}{2}, \frac{1}{2})$ and $Z^-(-\frac{1}{2}, -\frac{1}{2})$ correspond to $v = +\infty$ and $v = -\infty$, i.e. the contributions of strips infinitely far on either side. The chord $Z^- Z^+$ represents the unobstructed wave; a chord from a point v_0 to Z^+ gives the amplitude behind a straight edge.

Section B · Paragraph / problem
5 marks each.

25 PYQ N24 Q17

Width of principal maxima, N slits.

KNOW FIRST

T7 Width of a principal maximum T6 N slits: the intensity distribution M2 Small angles, limits and units

The N-slit intensity is $I = I_0 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma}$, $\gamma = \frac{\pi d \sin\theta}{\lambda}$.

1. m-th principal maximum: $\gamma = m\pi$, i.e. $d \sin\theta_m = m\lambda$
2. $\sin N\gamma = 0$ there, and the nearest zeros on either side are at $N\gamma = Nm\pi \pm \pi$, i.e. $d \sin(\theta_m \pm \Delta\theta_m) = m\lambda \pm \frac{\lambda}{N}$
3. Expand: $\sin(\theta_m \pm \Delta\theta_m) \approx \sin\theta_m \pm \Delta\theta_m \cos\theta_m$ (Δθ small)
4. Subtract: $d \cos\theta_m \, \Delta\theta_m = \frac{\lambda}{N}$

ANSWER $\Delta\theta_m = \frac{\lambda}{N d \cos\theta_m}$ **(half-width); the maxima get sharper as N increases**

Marks: intensity expression 1 · condition for zeros 2 · approximation and result 2.

26

Two-slit intensity; d = 3b.

KNOW FIRST

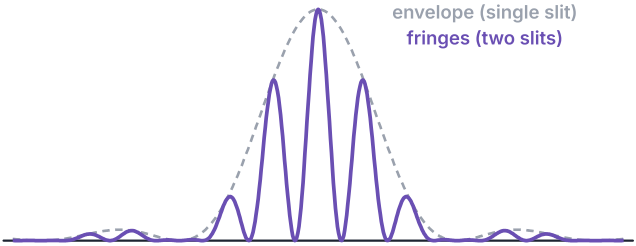
T5 Two slits: fringes inside an envelope T2 The single slit: intensity formula M1 Adding many arrows (phasors)

1. Each slit of width b gives at angle θ the field $A\frac{\sin\beta}{\beta}\cos(\omega t-\beta)$, $\beta=\frac{\pi b\sin\theta}{\lambda}$

2. The second slit, centre d away, adds a phase $\Phi=\frac{2\pi}{\lambda}d\sin\theta=2\gamma$

3. $E=A\frac{\sin\beta}{\beta}[\cos(\omega t-\beta)+\cos(\omega t-\beta-2\gamma)]=$ $2A\frac{\sin\beta}{\beta}\cos\gamma\,\cos(\omega t-\beta-\gamma)$ $\cos P+\cos Q=2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}.$

4. $I=4I_0\frac{\sin^2\beta}{\beta^2}\cos^2\gamma$



here d = 3b: the 3rd, 6th ... bright fringes are missing

Two slits, each of width b, centres d apart: $I = 4I_0\left(\frac{\sin\beta}{\beta}\right)^2\cos^2\gamma$. Fine interference fringes ($\cos^2\gamma$) sit inside the single-slit envelope. Where a fringe falls on an envelope zero, it is a **missing order**.

d = 3b: maxima $d\sin\theta = m\lambda$ coincide with envelope zeros $b\sin\theta = p\lambda$ when $m = 3p$: orders **3, 6, 9, ...** are missing. The central envelope ($|b\sin\theta| < \lambda$, i.e. $|d\sin\theta| < 3\lambda$) contains orders 0, ±1, ±2: **5 fringes**.

27 PYQ N22 Q29

Resolving power of a grating; numerical.

KNOW FIRST

T9 Resolving power of a grating T7 Width of a principal maximum

1. m-th order maximum of λ: $d\sin\theta = m\lambda$; of $\lambda + d\lambda$: $d\sin\theta' = m(\lambda + d\lambda)$
2. First minimum next to the maximum of λ: $d\sin\theta = m\lambda + \frac{\lambda}{N}$

3. Rayleigh: just resolved if these coincide: $m(\lambda + d\lambda) = m\lambda + \frac{\lambda}{N}$, so $m\,d\lambda = \frac{\lambda}{N}$

4. $R = \frac{\lambda}{d\lambda} = mN$

Numerical: $N = 0.15 \times 6 \times 10^5 = 9 \times 10^4$; first order $R = 1 \times 90000$.

ANSWER R = 90 000

> **SAME IDEA: N25 Q15**
>
> 1 inch with 15000 lines, first order, shortest λ = 500 nm: $R = 15000$, $d\lambda = \frac{500}{15000} = 0.033$ nm. The next resolvable wavelength is **500.033 nm**.

28 PYQ N20 Q27

Separation of sodium D lines.

KNOW FIRST

T8 The diffraction grating M2 Small angles, limits and units

1. $d = \frac{2.54}{15000} = 1.693 \times 10^{-4}$ cm

2. First order: $\sin\theta = \frac{5.89 \times 10^{-5}}{1.693 \times 10^{-4}} = 0.348$, $\cos\theta = 0.938$

3. $\Delta\theta = \frac{m\,\Delta\lambda}{d\cos\theta} = \frac{6 \times 10^{-8}}{1.693 \times 10^{-4} \times 0.938} = 3.78 \times 10^{-4}$ rad

4 Separation in the focal plane: $f\Delta\theta = 10 \times 3.78 \times 10^{-4} = 3.78 \times 10^{-3}$ cm

ANSWER ≈ 0.038 mm in the first order (≈ 0.099 mm in the second order)

29 PYQ N19 Q19

Resolving power of a microscope; oil immersion.

KNOW FIRST

T4 A round hole, and the limit of resolution T9 Resolving power of a grating

The resolving power of a microscope is its ability to show two close points of an object as separate. Because of diffraction at the objective, each point gives an Airy disc; by Rayleigh's criterion two points are just resolved when the centre of one disc falls on the first dark ring of the other. This gives the smallest separable distance

$$d_{min} = \frac{0.61\lambda}{n \sin \alpha}$$

where α is the semi-angle of the cone of light entering the objective and n the refractive index of the medium between object and objective; $n \sin \alpha$ is the numerical aperture (NA). Resolving power $\propto 1/d_{min} = \frac{n \sin \alpha}{0.61\lambda}$.

Oil immersion: filling the gap with cedar-wood oil ($n \approx 1.5$, nearly equal to glass) increases NA by about 1.5 times, so d_{min} decreases and finer detail is resolved. It also reduces light loss by reflection

at the cover slip. (Using shorter wavelength, e.g. UV, also improves resolution.)

30 PYQ N19 Q28

Resolution of the eye.

KNOW FIRST

T4 A round hole, and the limit of resolution

1. $\theta_{min} = \dfrac{1.22\lambda}{D} = \dfrac{1.22 \times 6 \times 10^{-7}}{2 \times 10^{-3}} = 3.66 \times 10^{-4}$ rad

2. Separation $x = L\theta = 30 \times 3.66 \times 10^{-4} = 1.10 \times 10^{-2}$ m

ANSWER About 1.1 cm

31 PYQ N20 Q21

Fraunhofer diffraction at a circular aperture.

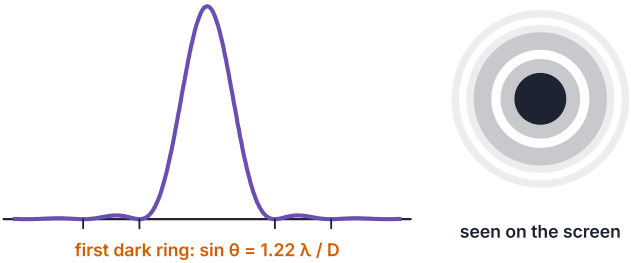
KNOW FIRST

T4 A round hole, and the limit of resolution T2 The single slit: intensity formula

A plane wave falls normally on a circular aperture of radius a and the pattern is observed in the focal plane of a lens. By symmetry the pattern consists of circular rings. Adding the contributions of all elements of the aperture gives

$$I = I_0 \left[\frac{2J_1(v)}{v}\right]^2, \qquad v = \frac{2\pi a}{\lambda}\sin\theta$$

where J_1 is the Bessel function of order one ($2J_1(v)/v \to 1$ as $v \to 0$).



Fraunhofer diffraction by a circular hole of diameter D: a bright central disc (the Airy disc, about 84% of the light) surrounded by faint rings. The first dark ring is at $\sin\theta = 1.22\lambda/D$.

- Central bright disc (Airy disc), containing about 84% of the light.
- Dark rings where $J_1(v) = 0$: $v = 3.832, 7.016, 10.17, \ldots$; the first at $\sin\theta = \dfrac{0.61\lambda}{a} = \dfrac{1.22\lambda}{D}$.
- Weak bright rings between them, of rapidly decreasing intensity.
- Radius of the first dark ring in the focal plane ≈ $\dfrac{1.22\lambda f}{D}$: smaller aperture → larger disc.

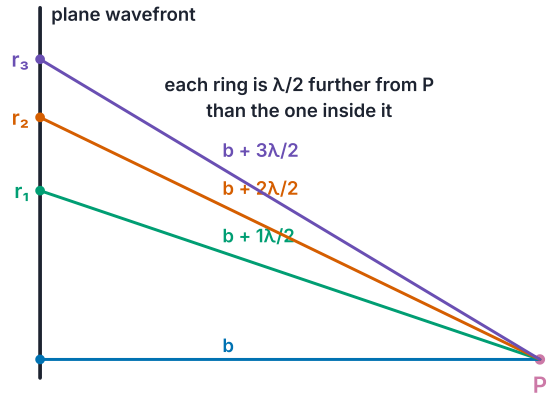
This sets the resolving limit of telescopes and microscopes. (N22 Q22 asks the same.)

32 PYQ N19 Q20

Construction of half-period zones; area.

KNOW FIRST

T10 Fresnel's half-period zones



Fresnel half-period zones. Divide the wavefront into rings whose distances to P increase by λ/2 each time. Light from neighbouring zones arrives half a wave apart (out of step), so their effects nearly cancel. $r_m \approx \sqrt{m\lambda b}$.

1. Let a plane wavefront WW′ travel towards P, at perpendicular distance $OP = b$. With P as centre, draw spheres of radii $b + \frac{\lambda}{2}, b + \frac{2\lambda}{2}, b + \frac{3\lambda}{2}, \ldots$, cutting the wavefront in circles of radii $r_1, r_2, r_3, \ldots$

2. The annular regions between successive circles are the half-period zones: light from corresponding points of adjacent zones reaches P with path difference λ/2 (half a period apart)

3. $r_n^2 = \left(b + \frac{n\lambda}{2}\right)^2 - b^2 = n\lambda b + \frac{n^2\lambda^2}{4} \approx n\lambda b$, so $r_n = \sqrt{n\lambda b}$

4. Area of n-th zone $= \pi(r_n^2 - r_{n-1}^2) = \pi\lambda b$

ANSWER **Area of each zone ≈ πλb: the same for all zones; radii ∝ √n**

Since the areas are equal, the amplitudes u_n depend only on the slowly increasing distance and obliquity, so they decrease slowly, and the resultant is $u_1/2$. (N24 Q19 asks the same with figure.)

33

Circular hole, radius 0.5 mm, λ = 5000 Å.

KNOW FIRST

T11 Straight-line travel, holes and discs T10 Fresnel's half-period zones

1. Number of zones in the hole as seen from an axial point at distance d: $p = \dfrac{a^2}{\lambda d}$
2. Maximum when p is odd; the farthest maximum is for p = 1: $d = \dfrac{a^2}{\lambda} = \dfrac{(0.05)^2}{5 \times 10^{-5}} = 50$ cm
3. Minimum when p is even; the farthest for p = 2: $d = \dfrac{a^2}{2\lambda} = 25$ cm

At 50 cm the hole contains only the first zone: amplitude u_1, twice the unobstructed $u_1/2$, so intensity 4 times. At 25 cm it contains two zones that nearly cancel: almost dark. Closer to the hole the maxima and minima alternate (p = 3, 4, ...: 16.7 cm, 12.5 cm, ...).

ANSWER Maximum at 50 cm; minimum at 25 cm

34 PYQ N22 Q21

Principle and construction of a zone plate.

KNOW FIRST

T12 The zone plate T10 Fresnel's half-period zones



every other zone blocked

only zones that add in step are left open → bright focus

$f = r_1^2 / \lambda$

also weaker foci at f/3, f/5, ...

A zone plate: rings with radii $\propto \sqrt{m}$, alternate rings blackened. All open zones send light to P in step, so P is very bright. It acts like a lens with focal length r_1^2/λ.

Construction: concentric circles with radii proportional to the square roots of the natural numbers ($r_n = r_1\sqrt{n}$) are drawn on white paper, and alternate annular zones are blackened. A reduced photograph on a glass plate gives the zone plate.

Principle: for a point P on the axis at distance $b = r_1^2/\lambda$, the rings are exactly the half-period zones. Without the plate, $u = u_1 - u_2 + u_3 - \cdots \approx u_1/2$. With the even zones blocked, $u = u_1 + u_3 + u_5 + \cdots$: all the transmitted light arrives in phase, so P

is extremely bright. The plate focuses light like a convex lens of focal length $f = r_1^2/\lambda$.

At $b = f/3$ each transparent ring contains three zones (net one), so there is another, weaker focus; similarly at $f/5, f/7, \ldots$.

35 PYQ N19 Q29

Foci of a zone plate, $r_n = 0.001\sqrt{n}$ m, λ = 4.5 × 10^{-7} m.

KNOW FIRST

T12 The zone plate

1. $r_1 = 0.001$ m, so $f = \dfrac{r_1^2}{\lambda} = \dfrac{10^{-6}}{4.5 \times 10^{-7}} = 2.22$ m
2. Other foci: $f/3 = 0.741$ m, $f/5 = 0.444$ m, $f/7 = 0.317$ m, ...

ANSWER 2.22 m, 0.741 m, 0.444 m, 0.317 m, ...

> **SAME IDEA: N20 Q30**
> $f = 60$ cm, λ = 6000 Å: $r_1 = \sqrt{f\lambda} = \sqrt{60 \times 6 \times 10^{-5}} = 0.06$ cm = **0.6 mm**.

36

Lens formula for a zone plate.

KNOW FIRST

T12 The zone plate

1. Point source S on the axis at distance a, image point P at distance b on the other side; L a point on the edge of the first zone, at height r_1

2. First-zone condition: $SL + LP - SP = \dfrac{\lambda}{2}$

3. $SL = \sqrt{a^2 + r_1^2} \approx a + \dfrac{r_1^2}{2a}$, $LP \approx b + \dfrac{r_1^2}{2b}$, $SP = a + b$

   Binomial approximation, $r_1 \ll a, b$.

4. $\dfrac{r_1^2}{2}\left(\dfrac{1}{a} + \dfrac{1}{b}\right) = \dfrac{\lambda}{2}$

5. $\dfrac{1}{a} + \dfrac{1}{b} = \dfrac{\lambda}{r_1^2} \equiv \dfrac{1}{f}$

ANSWER $\dfrac{1}{a} + \dfrac{1}{b} = \dfrac{1}{f}$ with $f = \dfrac{r_1^2}{\lambda} = \dfrac{r_n^2}{n\lambda}$

Since $f \propto 1/\lambda$, the zone plate has large chromatic aberration (red focused nearer than violet).

37

Straight edge: first two maxima.

KNOW FIRST

T13 The straight edge and the Cornu spiral

1. Distance from the shadow edge: $y = |v_0|\sqrt{\dfrac{\lambda d}{2}}$

2. $\sqrt{\dfrac{6 \times 10^{-5} \times 120}{2}} = \sqrt{3.6 \times 10^{-3}} = 0.06$ cm

3. $y_1 = 1.22 \times 0.06 = 0.073$ cm; $y_2 = 2.34 \times 0.06 = 0.140$ cm

ANSWER 0.73 mm and 1.40 mm into the illuminated region

38

Single slit 0.2 mm, 5000 Å, f = 20 cm.

KNOW FIRST

T3 Single slit: dark bands, weak bright bands, width

M2 Small angles, limits and units

1. Minima: $b \sin\theta = m\lambda$, small angles: $\theta_m = \dfrac{m\lambda}{b}$
2. $\dfrac{\lambda}{b} = \dfrac{5 \times 10^{-5}}{2 \times 10^{-2}} = 2.5 \times 10^{-3}$ rad
3. First minimum: $x_1 = f\theta_1 = 20 \times 2.5 \times 10^{-3} = 0.05$ cm; second: $x_2 = 0.10$ cm (each side of the centre)
4. Width of central maximum $= 2x_1 = 0.10$ cm

ANSWER Minima at 0.5 mm and 1.0 mm from the centre; central maximum 1.0 mm wide

Section C · Essays
10 marks each.

39 PYQ N20 Q33

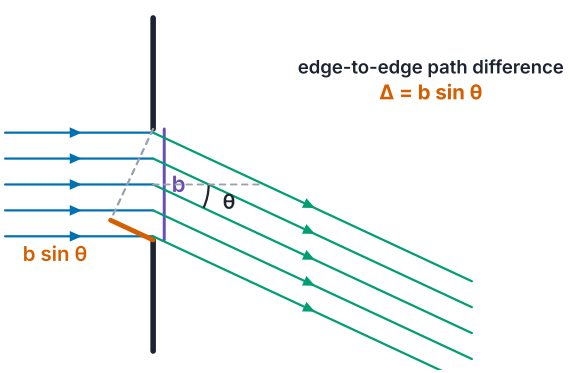
Fraunhofer diffraction by a single slit: maxima, minima, intensity diagram.

KNOW FIRST

T2 The single slit: intensity formula T3 Single slit: dark bands, weak bright bands, width T1 Diffraction; Fresnel and Fraunhofer M1 Adding many arrows (phasors) M4 Solving tan β = β with a graph

1. Arrangement

Monochromatic light from a slit source S at the focus of lens L_1 becomes a plane wave falling normally on a narrow slit AB of width b. Lens L_2 collects the light diffracted at each angle θ to a point P of its focal plane (the screen).



Single slit of width b, light diffracted at angle θ. The rays from the two edges differ in path by $b\sin\theta$. When this is one whole wavelength, the slit splits into two halves that cancel pair by pair: the first dark band.

2. Theory

1. Divide the slit into n equal strips of width Δ ($b = n\Delta$), each a source of amplitude a
2. Path difference between neighbouring strips in direction θ: $\Delta \sin\theta$; phase difference $\phi = \frac{2\pi}{\lambda}\Delta \sin\theta$
3. Resultant (sum of n equal vectors each turned by φ): $E_\theta = a\frac{\sin(n\phi/2)}{\sin(\phi/2)}$
4. Let $n \to \infty$, $na = A$: $\frac{n\phi}{2} = \frac{\pi b \sin\theta}{\lambda} = \beta$, $\sin\frac{\phi}{2} \approx \frac{\beta}{n}$, so $E_\theta = A\frac{\sin\beta}{\beta}$
5. $I = I_0\frac{\sin^2\beta}{\beta^2}$, $\beta = \frac{\pi b \sin\theta}{\lambda}$

3. Principal maximum

At θ = 0, β = 0 and $\frac{\sin\beta}{\beta} \to 1$: $I = I_0$, the central (principal) maximum. All wavelets arrive in phase.

4. Minima

$I = 0$ when $\sin\beta = 0$, β ≠ 0: $\beta = \pm m\pi$, i.e.

$$b\sin\theta = \pm m\lambda, \qquad m = 1, 2, 3, \ldots$$

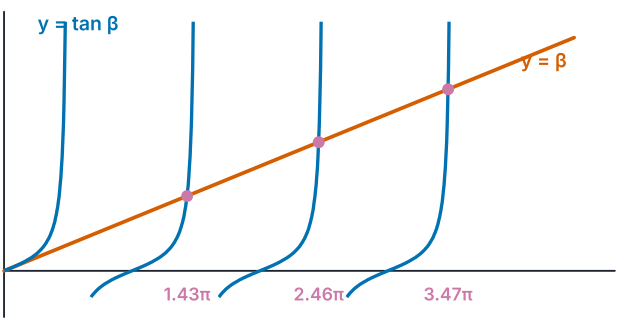
(Physically, at $b\sin\theta = \lambda$ the slit divides into two halves whose corresponding wavelets differ by λ/2 and cancel in pairs.)

5. Secondary maxima

1. $\dfrac{dI}{d\beta} = I_0 \dfrac{2\sin\beta(\beta\cos\beta - \sin\beta)}{\beta^3} = 0$

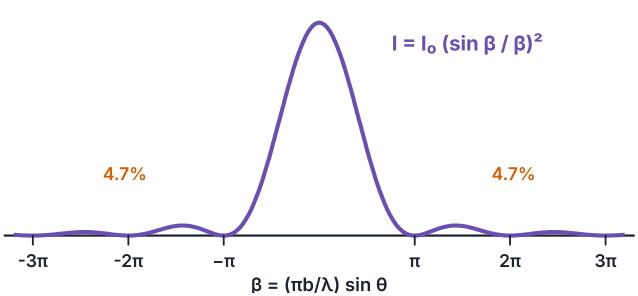
2. $\sin\beta \neq 0 \Rightarrow \tan\beta = \beta$; graphical solution: $\beta = \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi, \ldots$

3. Intensities: $I_0 \left(\dfrac{\sin 1.43\pi}{1.43\pi}\right)^2 = 0.047 I_0$, then $0.016 I_0$, $0.008 I_0$



The secondary maxima of a single slit are where the line $y = \beta$ meets $y = \tan\beta$ (pink dots): at $\beta = 1.43\pi, 2.46\pi, \ldots$, a little before $1.5\pi, 2.5\pi, \ldots$.

6. Intensity distribution



Single-slit Fraunhofer pattern. A bright, wide central maximum; dark points at $\beta = \pm\pi, \pm 2\pi, \ldots$ ($b\sin\theta = m\lambda$); weak secondary maxima (the first is only 4.7% of the centre).

The pattern is a bright central band of angular half-width λ/b (twice as wide as the others), flanked by much weaker secondary bands, symmetric about the centre, with intensities in the ratio $1 : \frac{1}{21} : \frac{1}{61} : \frac{1}{121}$ approximately. The width on the screen is $2f\lambda/b$: a narrower slit or a longer wavelength gives a wider pattern.

Marks: arrangement and figure 2 · derivation of intensity 3 · minima 1.5 · secondary maxima 2 · intensity graph and description 1.5.

Same essay: N19 Q33, N22 Q33, N24 Q21.

40 PYQ N25 Q21

Intensity distribution for N identical slits.

KNOW FIRST

T6 N slits: the intensity distribution T2 The single slit: intensity formula T5 Two slits: fringes inside an envelope M1 Adding many arrows (phasors) M3 Complex numbers as arrows

1. Set-up

N parallel slits, each of width b, with spacing (grating element) d, illuminated normally by a plane wave; diffracted light at angle θ is focused by a lens.

2. Field from each slit

Each slit alone gives $A\frac{\sin\beta}{\beta}\cos(\omega t - \beta)$ with $\beta = \frac{\pi b \sin\theta}{\lambda}$. Corresponding points of neighbouring

slits differ in path by $d\sin\theta$, phase $\Phi = \frac{2\pi d\sin\theta}{\lambda} = 2\gamma$.

3. Summing

1. $E = A\frac{\sin\beta}{\beta}\sum_{j=0}^{N-1}\cos(\omega t - \beta - 2j\gamma)$
2. The sum of N equal vectors each turned by 2γ has amplitude $\frac{\sin N\gamma}{\sin\gamma}$ (polygon of vectors / geometric series)
3. $E = A\frac{\sin\beta}{\beta}\frac{\sin N\gamma}{\sin\gamma}\cos[\omega t - \beta - (N-1)\gamma]$
4. $I = I_0\frac{\sin^2\beta}{\beta^2}\cdot\frac{\sin^2 N\gamma}{\sin^2\gamma}$

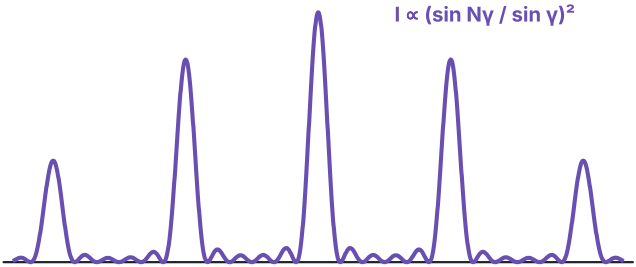
The first factor is the single-slit diffraction envelope; the second is the interference of N waves.

4. Principal maxima

When $\gamma = m\pi$ ($d\sin\theta = m\lambda$), $\frac{\sin N\gamma}{\sin\gamma} \to \pm N$, so $I = N^2 I_0\frac{\sin^2\beta}{\beta^2}$: very intense maxima, whose positions do not depend on N.

5. Minima and secondary maxima

$I = 0$ when $\sin N\gamma = 0$ but $\sin\gamma \neq 0$: $N\gamma = p\pi$, $p \neq 0, N, 2N, \ldots$, i.e. $d\sin\theta = \frac{\lambda}{N}, \frac{2\lambda}{N}, \ldots, \frac{(N-1)\lambda}{N}, \frac{(N+1)\lambda}{N}, \ldots$. Between adjacent principal maxima there are $N-1$ minima and $N-2$ feeble secondary maxima.



N = 6: sharp principal maxima, N − 2 = 4 tiny ones between

N slits. The principal maxima stay at the same places ($d\sin\theta = m\lambda$) but become taller ($\propto N^2$) and much narrower as N grows. Between two principal maxima there are $N-1$ minima and $N-2$ weak secondary maxima.

6. Width and missing orders

The half-width of a principal maximum is $\dfrac{\lambda}{Nd\cos\theta}$: it becomes sharper as N increases. A principal maximum is missing if it falls at a zero of the envelope, $b\sin\theta = p\lambda$, i.e. if $d/b = m/p$.

7. Special cases

N = 1: single slit. N = 2: $\dfrac{\sin 2\gamma}{\sin\gamma} = 2\cos\gamma$, giving $4I_0\dfrac{\sin^2\beta}{\beta^2}\cos^2\gamma$ (double slit). Large N: diffraction grating.

Marks: set-up 1 · summation and intensity 4 · principal maxima 2 · minima/secondary maxima 2 · graph 1.

41 PYQ N22 Q34

Rectilinear propagation from half-period zones.

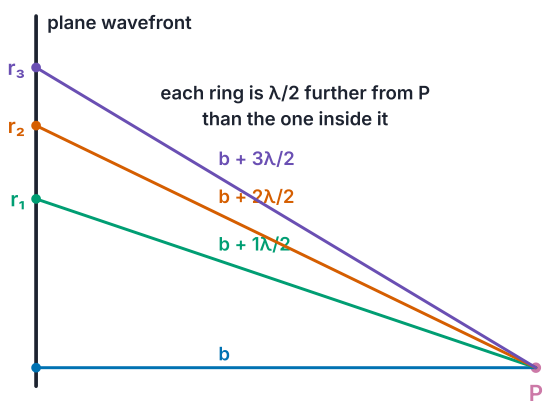
KNOW FIRST

T10 Fresnel's half-period zones T11 Straight-line travel, holes and discs

1. The problem

Light is a wave, yet it appears to travel in straight lines and to cast sharp shadows. Fresnel explained this with half-period zones.

2. Half-period zones



Fresnel half-period zones. Divide the wavefront into rings whose distances to P increase by λ/2 each time. Light from neighbouring zones arrives half a wave apart (out of step), so their effects nearly cancel. $r_m \approx \sqrt{m\lambda b}$.

For a plane wavefront and a point P at distance b, spheres of radii $b + n\lambda/2$ about P divide the wavefront into zones of radii $r_n = \sqrt{n\lambda b}$ and equal areas $\pi\lambda b$. Waves from adjacent zones reach P with phase difference π.

3. Resultant at P

1. $u = u_1 - u_2 + u_3 - u_4 + \cdots$, with u_n slowly decreasing (obliquity and distance)
2. $u = \frac{u_1}{2} + \left(\frac{u_1}{2} - u_2 + \frac{u_3}{2}\right) + \left(\frac{u_3}{2} - u_4 + \frac{u_5}{2}\right) + \cdots \approx \frac{u_1}{2}$

4. Rectilinear propagation

The effect at P is thus half that of the first zone. The outer zones cancel each other in pairs; effectively only a small region around the foot O of the perpendicular from P matters. For $b = 50$ cm and λ = 5×10^{-5} cm, $r_1 = \sqrt{\lambda b} = 0.05$ cm: the light reaching P comes, in effect, from a disc only half a millimetre across around the straight line OP. So light appears to travel along the straight line from the source to P.

5. Shadows

If an obstacle much larger than the first few zones is placed between the source and P, covering the region around OP, P receives practically no light: a sharp shadow. If P lies in the illuminated region, far from the edge of the obstacle, the first few zones around OP are unobstructed and P receives the normal intensity $u_1/2$. Only near the edge of the geometrical shadow, within a few zone-widths ($\sqrt{\lambda b}$, a fraction of a millimetre), does the light deviate: there diffraction fringes appear.

6. When it fails

If the obstacle or aperture is comparable with the zone size (very small holes, discs, narrow slits), the straight-line picture fails: a small hole gives a bright or dark centre depending on the number of zones it exposes, and a small disc gives a bright spot at the centre of its shadow.

7. Conclusion

Because λ is so small, the effective zones are tiny, so in everyday conditions light behaves as if it travels in straight lines. Rectilinear propagation is thus a consequence of the wave theory.

Marks: zone construction and radii/areas 3 · resultant $u_1/2$ 3 · rectilinear propagation and shadows 3 · conclusion 1.

42

Fresnel diffraction at a straight edge; Fresnel integrals; Cornu spiral.

KNOW FIRST

T13 The straight edge and the Cornu spiral M3 Complex numbers as arrows T10 Fresnel's half-period zones

1. Arrangement

A plane wave falls on an opaque straight edge (the edge along the ξ-axis, blocking one side). A screen at distance d is observed at a point P at height y from the edge of the geometrical shadow.

2. Amplitude at P

1. Distance from a point (ξ, η) of the open wavefront to P: $r \approx d + \dfrac{\xi^2 + (\eta - y)^2}{2d}$

2. Summing the wavelets: $u(P) \propto \displaystyle\int_{-\infty}^{\infty} d\xi \int_{0}^{\infty} d\eta \, e^{ik[\xi^2 + (\eta - y)^2]/2d}$

3. Substitute $u = \sqrt{\dfrac{2}{\lambda d}}\,\xi$, $v = \sqrt{\dfrac{2}{\lambda d}}(\eta - y)$; the exponent becomes $\dfrac{i\pi}{2}(u^2 + v^2)$, and the lower limit of v is $v_0 = -\sqrt{\dfrac{2}{\lambda d}}\,y$

4. $u(P) = \dfrac{1 - i}{2}\, u_0 \left\{ \left[\tfrac{1}{2} - C(v_0)\right] + i \left[\tfrac{1}{2} - S(v_0)\right] \right\}$, with $C(v) = \int_0^v \cos \dfrac{\pi t^2}{2} dt$, $S(v) = \int_0^v \sin \dfrac{\pi t^2}{2} dt$

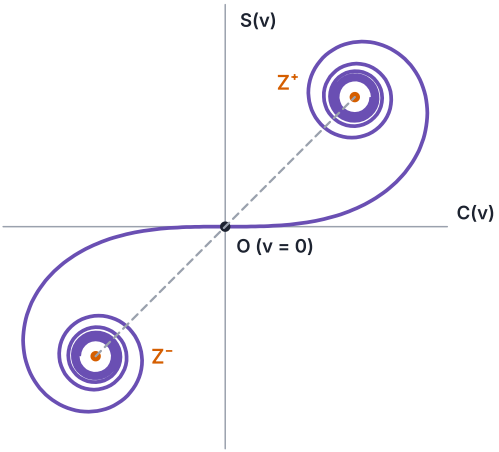
3. Intensity

Since $\left| \dfrac{1 - i}{2} \right|^2 = \dfrac{1}{2}$:

$$I(P) = \frac{I_0}{2} \left\{ \left[\tfrac{1}{2} - C(v_0)\right]^2 + \left[\tfrac{1}{2} - S(v_0)\right]^2 \right\}$$

where I_0 is the unobstructed intensity (check: $v_0 \to -\infty$, $C, S \to -\dfrac{1}{2}$, giving $I = I_0$).

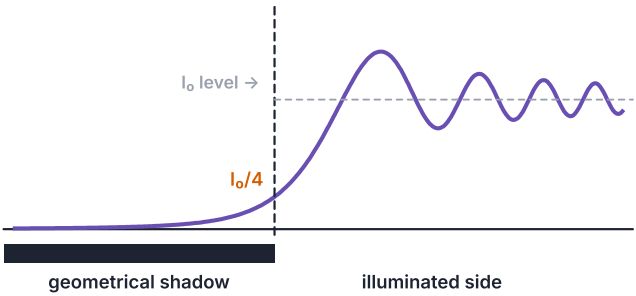
4. Cornu spiral



The Cornu spiral: the points $(C(v), S(v))$ of the Fresnel integrals. The amplitude at a point of the screen is the length of the chord between the right two points of the spiral. Intensity ∝ (chord)².

The points $(C(v), S(v))$ form the Cornu spiral, winding into $Z^+(\frac{1}{2}, \frac{1}{2})$ and $Z^-(-\frac{1}{2}, -\frac{1}{2})$. The amplitude at P is proportional to the length of the chord from the spiral point v_0 to Z^+, and the intensity to its square.

5. The pattern



Diffraction at a straight edge. In the shadow the light fades smoothly to zero; at the edge of the shadow it is exactly $I_0/4$; on the bright side it wobbles above and below I_0 (first peak ≈ 1.37 I_0) before settling.

- **Edge of the shadow** ($y = 0$, $v_0 = 0$): chord OZ⁺ is half of Z⁻Z⁺, so $I = I_0/4$.
- **Geometrical shadow** ($v_0 > 0$): as v_0 increases, the point moves into the upper curl and the chord shrinks steadily: the intensity falls smoothly to zero, without fringes.
- **Illuminated region** ($v_0 < 0$): the point moves round the lower curl, so the chord alternately grows and shrinks: fringes. Maxima at $|v_0| = 1.22, 2.34, 3.08$ ($1.37I_0, 1.20I_0, 1.15I_0$), minima at 1.87, 2.74, … ($0.78I_0$, $0.84I_0$). They get closer together and the intensity settles at I_0.
- Positions on the screen: $y = |v_0|\sqrt{\lambda d/2}$.

Marks: set-up 1 · integrals and amplitude 3 · Cornu spiral 2 · intensity at edge, shadow and lit region with graph 4.